

Original Article

Healthcare-associated infections in Italian long-term care facilities: a machine learning analysis of a 12-month cohort

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Abstract

Objectives: To estimate the incidence of healthcare-associated infections (HAIs) in Italian long-term care facilities (LTCFs) and to evaluate whether an artificial intelligence (AI) approach, through unsupervised machine learning (ML), could stratify residents into clinically distinct groups with differing susceptibility to HAIs.

Design: Prospective cohort study with 12-month follow-up.

Setting: 24 LTCFs in Italy, participating in the European Centre for Disease Prevention and Control 12-month longitudinal study on HAIs in LTCFs, 2022–2023.

Participants: 395 residents enrolled across the participating LTCFs.

Methods: Incidence measures of HAIs (rate and ratio) were estimated, using generalized estimating equations. A hierarchical cluster analysis based on residents' clinical and demographic characteristics was implemented as an unsupervised ML approach.

Results: Overall, 75 HAIs per 100 residents (95% CI, 70.3–78.3) and 0.23 HAIs per 1,000 resident-days (95% CI, 0.11–0.76) were estimated. Respiratory tract infections (29.5%, 95% CI 24.2–31.1), COVID-19 (26.3%, 95% CI 22.1–28.4), and urinary tract infections (15%, 95% CI 11.0–35.4) were the most frequent. Clustering identified two reproducible resident groups: Group 1 (39%), more independent and cognitively preserved, with fewer comorbidities and lower infection incidence; and Group 2 (61%), more dependent and clinically complex, with higher incidence of HAIs. Cluster stability was high (mean ARI = 0.83).

Conclusions: This study confirms the high burden of HAIs in Italian LTCFs and provides exploratory evidence that AI-based clustering can identify reproducible HAI susceptibility profiles in a setting where such approaches have been scarcely applied.

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Introduction

More than 3.5 million Healthcare-Associated Infections (HAIs) are reported in the European Union/European Economic Area (EU/EEA) each year, resulting in more than 90,000 deaths and approximately 2.5 million disability-adjusted life years (DALYs).¹ The incidence of HAIs with adverse patient outcomes increases with age, making older people more vulnerable to HAIs and their complications.² Long-term care facilities (LTCFs), due to their community-based nature and frail elderly users, are particularly exposed to increased incidences of HAI-associated morbidity and mortality.

The Healthcare associated infections and antimicrobial use in European LTCFs projects, led by the European Centre for Disease Prevention and Control (ECDC), reported a pooled HAI prevalence between 2.6% and 3.7%.^{3,4} Based on national prospective

surveillance studies, the incidence of HAIs in European ranges from 2.1 to 11.8 per 1,000 resident days.^{5,6} A pilot Point Prevalence Study conducted in 2022 in Italy, found that 2.5% of residents experienced at least one case of HAI, including COVID-19.⁷ A previous nationwide Italian study estimated that there were approximately 641,065 new HAI cases and 29,375 attributable deaths over a one-year period and the total annual burden was calculated to be 702.5 DALYs per 100,000 inhabitants.⁸ Urinary tract infections (UTIs), respiratory tract infections (RTIs), gastrointestinal tract infections and skin and soft tissues infections are among the most reported HAIs.⁹ A recent European 12-month longitudinal study led by ECDC in 2022–2023 (H4LS study, “Healthcare-associated infections and antimicrobial use in LTCFs—support to a point prevalence survey and a longitudinal study,”)¹⁰ found a high incidence of HAIs in 65 LTCFs. One in two residents had at least one HAI, leading to hospitalization in 4.3% of total HAIs and death in 4.5% of cases, with RTIs and UTIs accounting for almost half of all HAIs.

LTCF residents are often viewed as a relatively homogeneous, highly frail group with multiple functional deficits. However, an

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individualized approach recognizes substantial heterogeneity in care needs, particularly regarding cognition and mobility, which are inversely related and major drivers of LTCF admission.^{11,12} The risk of acquiring a HAI, with its associated increased morbidity and mortality, is influenced by specific resident characteristics including advanced age, underlying medical conditions, impaired cognitive and functional status, and the use of invasive devices such as indwelling urinary catheters.^{2,3} In this context, rather than quantifying frailty per se, identifying multidimensional resident profiles based on the concurrent assessment of functional, cognitive, and clinical characteristics associated with differential susceptibility to specific types of HAIs becomes fundamental.

Recent applications of machine learning (ML) models have demonstrated the potential to significantly enhance the prediction of HAIs. Predictive algorithms have been successfully applied to estimate the risk of HAIs in intensive care units,¹³ to identify bloodstream infections at an early stage,¹⁴ and to recognize patients at risk of acquiring RTIs, including COVID-19, during hospitalization.^{15,16} In many cases, these models have achieved promising predictive performance, with possible applications in supporting clinical decision-making and infection prevention and control (IPC).¹⁷ However, while artificial intelligence (AI) has already been successfully applied in hospital settings for HAI prediction and antimicrobial stewardship, its use in LTCFs is still largely unexplored.¹⁸ This study addressed this gap by applying an unsupervised ML approach in on LTCF resident data from the H4LS study.¹⁰ The aim was to identify resident subgroups with differing risk profiles and HAI susceptibility patterns based on a multidimensional clinical assessment, thereby supporting more targeted and personalized IPC strategies.

Methods

Study population and data collection procedures

This research was conducted following the guidelines outlined in the Strengthening the Reporting of Observational Studies in Epidemiology statement,¹⁹ using data collected within the H4LS study, aimed at assessing the incidence of HAIs and their associated mortality and hospitalizations in European LTCFs.¹⁰ Informed consent requirements prevented full baseline resident enrollment; therefore, a convenience sampling strategy was adopted rather than a complete census. Eligible residents were those expected to stay in the facilities for at least one year. The population of the present study consisted of 395 residents living in 24 LTCFs (nursing homes, residential homes or mixed type facilities), from two Italian regions, Emilia-Romagna and Piedmont.

The Italian study's National Survey Coordinator provided each LTCF with the survey protocol and the following operational tools: an Institutional Questionnaire to collect LTCF characteristics, a Resident Questionnaire to collect clinical and demographic information for each resident and a HAI Questionnaire to record each HAI case occurred during the follow-up period. Information on temporary discharges from the facilities were also collected, to include only the HAIs acquired within the LTCFs in the analyses. All data converged into a national study database. The H4LS study procedures are detailed elsewhere.¹⁰

Definition of HAI

To ensure unambiguous identification of HAIs, definitions of HAIs followed the McGeer and revised McGeer criteria as adopted in the H4LS study¹⁰ and described in the ECDC protocol for

surveillance in European LTCFs.²⁰ No modifications to these definitions were applied in the present analysis. The COVID-19 case definition was based on the positive result of a laboratory test. Asymptomatic cases were not considered. In addition to HAIs acquired within the facilities, HAIs related to a temporary discharge were also included when symptoms occurred more than two days after readmission, since these cases were partially or fully managed and treated within the LTCF.

Statistical analyses

The following HAI incidence metrics were computed from the crude sample data: (i) percentage, ie the number of HAIs per type, over the total number of HAIs; (ii) ratio, ie the cumulative incidence of HAIs per 100 residents, calculated as the number of HAIs per type divided by the total number of LTCF residents multiplied by 100; (iii) rate, ie the number of HAIs per 1,000 resident days, calculated as the ratio between the total number of HAIs and the sum of follow-up days multiplied by 1,000.

Due to the multi-center nature of the data, Generalized Estimating Equations models were also estimated with an appropriate link function (Poisson or binomial, based on variable types). The intercept of these models was used to obtain a clean measure of the multi-center effects.

Unsupervised ML was used to identify groups of residents with similar clinical-demographic characteristics, and a hierarchical cluster analysis was performed using a dissimilarity matrix based on the Gower metric and the complete linkage clustering method. Table 1 reports the clinical and demographic variables that were used for clustering. To assess the robustness of the clustering solution, a bootstrap resampling procedure with 1,000 iterations was implemented. The stability of the resulting solutions was quantified using the adjusted Rand index (ARI) between each resampled clustering and the original solution, and the mean ARI was calculated as an overall measure of cluster reproducibility. After clustering, the characteristics of the resulting groups were examined through descriptive statistics and the incidence of each type of HAI was calculated for each group by means of the following measures: (i) number of each type of HAI divided by the total number of HAIs in the group multiplied by 100; (ii) number of each type of HAI divided by the total number of residents in the group multiplied by 100. In addition, a *t*-test for quantitative variables and a χ^2 test for qualitative variables were performed to test for any significant differences between the groups. In this study, clustering was applied as a data-driven stratification tool to group residents into distinct multidimensional susceptibility profiles rather than to rank residents by overall frailty; comparisons with a Frailty Index²² and the Charlson Comorbidity Index²¹ are reported in Appendix A1.

Results

Clinical-demographic characteristics and incidence of HAIs in the resident population

The 395 residents in the study population were predominantly women (70%) and had a mean age of 84.6 years (SD 3.1). Their clinical characteristics were heterogeneous, with multiple health conditions (mean CCI = 2.9, SD: 2.2) (Table 2).

The total number of HAIs recorded during the study period was 296, with RTIs being the most reported (29.5%, 95% CI 24.2%–31.1%), followed by COVID-19 infections (26.3%, 95% CI 22.1%–28.4%) and UTIs (15%, 95% CI 11.0%–35.4%). Table 3 shows crude and estimated incidence measures by type of HAI.

Table 1. Demographic, functional, clinical and comorbidity variables included in the machine-learning clustering analysis

Demographic characteristics	
	Sex
	Age
Clinical characteristics	
	Disorientation
	Mobility
	Incontinence
	Urinary catheter
	Vascular catheter
Charlson Comorbidity Index conditions ¹	
	Myocardial infarction
	Decompensatio cordis
	Peripheral arterial disease
	Cardiovascular disease
	Dementia
	Chronic lung disease
	Peptic ulcer
	Liver disorder
	Hemiplegia
	Kidney disorder
	Chronic urinary disorder
	Malignancies
	Leukemia
	Lymphoma
	Metastases
	Systemic disease ²
	Diabetes

¹[21].²Systemic lupus erythematosus, polymyositis, mixed connective tissue disease, polymyalgia rheumatica, or moderate-to-severe rheumatoid arthritis.

Clustering of residents

The algorithm produced two significant groups, characterized by low intra-group heterogeneity and high inter-group heterogeneity, reflecting the different clinical case mix of the residents. The quality of the clustering solution was good (silhouette coefficient: 0.59) and reproducibility of clusters was high (ARI: 0.83). Group 1 (G1) included 156 residents (39% of the sample); Group 2 (G2) included 239 residents (61%) (Table 2).

Compared to G1, residents in G2 were predominantly women (G1: 64% vs G2: 75%, $P = .01$), disoriented (G1: 40% vs G2: 78%, $P < .0001$), in a wheelchair (G1: 29% vs G2: 86%, $P < .0001$) and incontinent (G1: 37% vs G2: 92%, $P < .0001$). Moreover, regarding the clinical characteristics, G2 included a higher percentage of residents with dementia (G1: 22% vs G2: 55%, $P = .001$), chronic lung disease (G1: 8% vs G2: 22%, $P < .0001$), hemiplegia (G1: 3% vs G2: 9%, $P = .02$), kidney disorders (G1: 7% vs G2: 31%, $P < .0001$), malignancies (G1: 5% vs G2: 11%, $P = .03$) and diabetes (G1: 14% vs G2: 29%, $P < .0001$). In contrast to G2, most of the G1 residents were ambulant (69.2%). G1 was characterized by a significantly higher percentage of residents with systemic disease (G1: 19.2% vs

G2: 5.0%, $P = .001$). The mean age of G1 and G2 was 83 (SD 8.9) and 85 (SD 10.6), respectively.

The proportion of HAIs in G2 was significantly higher than in G1 (G1: 65.4% vs. G2: 81.2%, $P < .0001$), highlighting an increased susceptibility of this group to HAIs, including UTIs (G1: 9.8% vs. G2: 19%, $P < .0001$), RTIs (G1: 27.5% vs. G2: 33.5%, $P = .04$) and skin and soft tissues infections (G1: 3.9% vs. G2: 8.8%, $P < .0001$). G1 was characterized by a higher percentage of COVID-19 infections (G1: 43.1% vs. G2: 20.1%, $P = .02$) (Table 4).

Figure 1 summarizes the clustering results shown in Tables 2 and 4.

Discussion

The longitudinal design of the study provides important evidence on the incidence of HAIs in a relatively large sample of Italian LTCFs. The estimated overall HAI incidence of 0.6 per 1,000 resident days complements incidence rates from earlier European studies, underscoring the need for targeted surveillance and prevention strategies.^{5,6}

Beyond its epidemiological contribution, this study extends the application of AI, particularly ML methods, from their predominant use in acute-care hospital settings,^{13–17} to LTCFs, suggesting their potential to predict HAIs and inform antimicrobial stewardship in this context as well.

Rather than relying solely on frailty, this method identified a highly stable two-cluster stratification, that captured multidimensional clinical profiles reflecting underlying health status and differential susceptibility to specific types of HAIs. The observed high level of cluster reproducibility indicates that the ML-derived resident stratification was robust and unlikely to result from sampling variability. Such stability reinforces the value of an ML-based approach for reliably defining HAI susceptibility profiles in LTCFs and supports its potential use as an early-warning component for IPC. In addition, the identification of high-risk clusters may enhance antimicrobial stewardship by prioritizing targeted screening and guiding differentiated antibiotic protocols, thereby reducing inappropriate antimicrobial use.

Compared to Group 1, Group 2 included more clinically vulnerable residents with higher exposure to RTIs, UTIs, and skin and soft tissues infections. These infections were significantly more frequent among residents with a high prevalence of underlying health conditions such as disorientation, incontinence, low mobility, hemiplegia, kidney disorder, dementia, malignancies, diabetes, and chronic lung disease. Most of these conditions are well-recognized risk factors for bacterial HAIs, particularly UTIs and skin or soft-tissue infections. The observed co-occurrence of incontinence and UTIs, and of chronic lung disease and RTIs aligns with previous reports and supports the biological plausibility of these findings.^{3,4,23,24} In contrast, Group 1 residents, characterized by better mobility, greater functional autonomy, preserved cognitive status, and by a lower overall comorbidity burden, showed a higher proportion of COVID-19 infections. Greater mobility within the facility and increased interpersonal contact may have facilitated exposure to respiratory viral pathogens, consistent with patterns described in LTCF COVID-19 outbreaks.²⁵ Notably, G1 included residents with clinically significant systemic diseases (as defined in Table 2), underscoring that serious inflammatory conditions may occur despite a relatively low overall comorbidity burden, reflecting substantial heterogeneity across individuals and among different rheumatic disorders.²⁶

Table 2. Demographic and clinical characteristics of LTCF residents, overall and by groups, after cluster analysis

	Total sample		G1		G2		G1 vs. G2 (p-values) ¹
Residents (N)	395		156		239		
Demographic and clinical characteristics							
<i>Sex</i>							
Female	278	70.4%	100	64.1%	178	74.5%	**
Age, years (mean, SD)	84.6	3.1	83.4	8.9	85.4	10.6	*
<i>Disorientation</i>							
Mild	106	26.8%	37	23.7%	69	28.9%	
Moderate	65	16.5%	12	7.7%	53	22.2%	***
Severe	77	19.5%	13	8.3%	64	26.8%	***
<i>Mobility</i>							
Ambulant	131	33.2%	108	69.2%	23	9.6%	***
Bedridden	14	3.5%	3	1.9%	11	4.6%	
Wheelchair	250	63.3%	45	28.6%	205	85.8%	***
Incontinence	279	70.6	57	36.5%	222	92.3%	***
Urinary catheter	24	6.1%	8	5.1%	16	6.7%	
Vascular catheter	7	1.8%	2	1.3%	5	2.1%	
Myocardial infarction	30	7.6%	12	7.7%	18	7.5%	
Decompensatio cordis	73	18.5%	28	18.0%	45	18.8%	
Peripheral arterial disease	52	13.2%	18	11.5%	34	14.2%	
Cardiovascular disease	135	34.2%	47	30.1%	88	36.8%	
Dementia	165	41.8%	34	21.8%	131	54.8%	***
Chronic lung disease	64	16.2%	12	7.7%	52	21.8%	***
Peptic ulcer	11	2.8%	5	3.2%	6	2.5%	
Liver disorder	14	3.5%	6	3.8%	8	3.3%	
Hemiplegia	25	6.3%	4	2.6%	21	8.8%	**
Kidney disorder	38	9.6%	7	4.5%	31	13.0%	***
Chronic urinary disorder	60	15.2%	22	14.1%	38	15.1%	
Malignancies	34	8.6%	8	5.1%	26	10.9%	**
Leukemia	3	0.7%	3	1.9%	0	0.0%	
Lymphoma	7	1.8%	3	1.9%		1.7%	
Metastases	7	1.8%	1	0.6%	6	2.5%	
Systemic disease	42	10.6%	30	19.2%	12	5.0%	***
Diabetes	92	23.3%	22	14.1%	70	9.3%	***
Charlson's index, CCI (mean, SD)	2.9	2.9	2.9	3.0	3.3	3.1	

¹P-values are reported as follows: * .01 ≤ P < .05; ** .001 ≤ P < .01; *** P < .001.

G1 and G2: groups resulting from the machine-learning clustering analysis.

Data source: Italian H4LS data set, 2023.

Prior studies have demonstrated that variation in resident frailty, disability, and clinical case mix substantially influences infection risk, prompting efforts to operationalize these differences using administrative measures such as the Case-Mix Index (CMI) and Resource Utilization Groups.^{27–30} However, no consistent association was observed between CMI and infection rates, with the notable exception of a study conducted in the same Italian regions included in the present analysis.³¹ Taken together, these findings suggest that traditional administrative measures may not

fully capture the multidimensional clinical vulnerability underlying susceptibility to HAIs in LTCFs. Likewise, frailty indices and comorbidity scores summarize vulnerability primarily as an overall burden of deficits, limiting their ability to differentiate distinct clinical profiles within already frail LTCF populations.^{32,33} In contrast, the unsupervised clustering approach adopted in this study offers a methodologically distinct and complementary perspective: rather than ranking residents along a single continuum of frailty or comorbidity, it integrates functional, cognitive, clinical,

Table 3. Crude percentage, estimated ratio and rate by type of HAIs in the total sample (n = 395)

HAIs	N	Crude %	Estimated %	Crude ratio	Estimated ratio (95% CI)	Crude rate	Estimated rate (95% CI)
Total infections	296	100	–	74.9	74.9 (70.3–78.3)	0.23	0.57 (0.11–0.76)
RTIs	93	31.4	29.5 (24.2–31.1)	24.5	24.5 (22.2–27.0)	0.07	0.94 (1.01–1.33)
COVID-19	83	28.0	26.3 (22.1–28.4)	22.7	22.7 (21.2–24.3)	0.07	0.55 (0.39–0.79)
UTIs	48	16.2	15.0 (11.0–35.4)	35.6	35.6 (33.2–38.2)	0.04	0.76 (0.60–0.95)
Gastrointestinal	22	7.4	7.6 (4.8–8.9)	7.8	7.8 (6.7–9.1)	0.02	0.23 (0.13–0.39)
Skin and soft tissues	21	7.1	6.9 (4.4–11.8)	8.7	8.7 (7.6–9.8)	0.02	0.59 (0.39–0.88)
Eye, ear, nose and mouth	17	5.7	5.7 (3.6–6.1)	3.6	3.6 (2.9–4.5)	0.01	0.22 (0.15–0.32)
Other	7	2.4	2.4 (1.1–20.2)	0.5	0.5 (0.3–0.8)	0.01	0.11 (0.08–0.16)
Surgical site	1	0.3	0.4 (0.1–10.7)	19.2	19.2 (17.2–21.4)	0.00	0.01 (0.01–0.03)
Unexplained febrile case	4	1.4	1.4 (1.1–4.9)	0.6	0.6 (0.3–0.9)	0.00	0.04 (0.02–0.07)
Bloodstream	0	0.0	–				

RTIs, respiratory tract infections; UTIs, urinary tract infections.
Data source: Italian H4LS data set, 2023.

Table 4. Percentage of type of HAIs by groups

HAIs		G1 (n = 156)	G2 (n = 239)	G1 vs G2 (p-values) ¹
Total infections	N	102	194	
	% on residents	65.4	81.2	***
UTIs	N	10	38	
	% on HAIs	9.8	19.6	***
	% on residents	6.6	8.0	***
RTIs	N	28	65	
	% on HAIs	27.5	33.5	*
	% on residents	18.5	13.7	***
COVID-19	N	44	39	
	% on HAIs	43.1	20.1	**
	% on residents	29.1	10.3	
Surgical site	N	1	0	
	% on HAIs	1.0	0.0	
	% on residents	0.7	0.0	
Skin and soft tissues	N	4	17	
	% on HAIs	3.9	8.8	***
	% on residents	2.6	3.6	.
Gastrointestinal tract	N	6	16	
	% on HAIs	5.9	8.2	
	% on residents	4.0	3.4	
Eye, ear, nose mouth	N	5	12	
	% on HAIs	4.9	6.2	
	% on residents	3.3	2.5	
Bloodstream	N	0	0	
	% on HAIs	0	0	

(Continued)

Table 4. (Continued)

HAIs		G1	G2	G1 vs G2 (p-values) ¹
	% on residents	0	0	
Unexplained febrile case	N	2	2	
	% on HAIs	2.0	1.0	
	% on residents	1.3	0.4	
Other	N	2	5	
	% on HAIs	2.0	2.6	
	% on residents	1.3	1.1	

RTIs, respiratory tract infections; UTIs, urinary tract infections; HAIs, healthcare-associated infections.

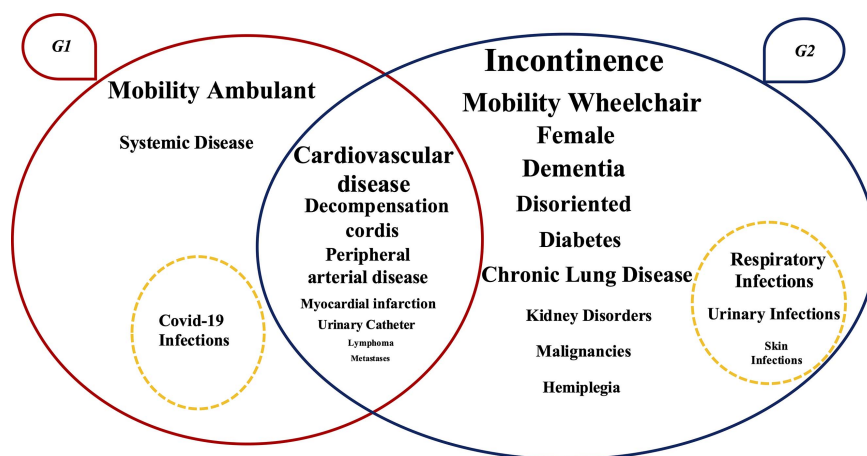
¹P-values are reported as follows: .10 < P < .05; * .01 ≤ P < .05; ** .001 ≤ P < .01; *** P < .001. Data source: Italian H4LS data set, 2023.

and care-related variables simultaneously, without imposing predefined weights or linear assumptions. This data-driven framework may facilitate the identification of latent clinical profiles defined by specific combinations of characteristics, thereby helping to capture heterogeneity that could be masked by uni-dimensional summary measures.

Consistent with this perspective, the two clusters identified in this study share several underlying health conditions, reflecting the generally high baseline vulnerability of the LTCF population. However, they differ in functional, cognitive, continence, and care-dependency patterns that may better capture infection susceptibility than overall frailty.

This approach aligns with contemporary models of geriatric care, which move beyond the traditional view of LTCF populations as uniformly frail, and instead adopt a more dynamic, risk-adjusted perspective.^{11,32} In this context, the integration of ML techniques into epidemiological surveillance enhances early warning capabilities and supports evidence-based clinical decision-making through resident-level data stratification. The well-documented severe consequences of HAIs among LTCF residents reinforce the

Figure 1. Visual representation of the clustering results (N = 395). Note: G1 and G2 are represented by ovals with the corresponding clinical conditions reported within them. The conditions positioned at the intersection of the ovals are those that do not show statistically significant differences between the two groups. Conversely, the conditions placed within each oval are those that differ significantly between the groups; each condition is reported in the group where it has a higher prevalence. The font size is proportional to the percentage of residents presenting each clinical condition. The dashed circle indicates the HAIs that are more prevalent in each group.



urgency of identifying high-risk resident groups and tailoring preventive interventions accordingly.³⁴ Additionally, high rates of colonization with multidrug-resistant organisms, as observed in Italian LTCFs,³⁵ further underscore the critical need for proactive surveillance and stratification strategies to mitigate infection risk.

Some limitations of this study should be considered when interpreting the findings. Incomplete resident enrollment due to informed consent requirements prevented a full census in some facilities, potentially limiting representativeness. Such barriers to comprehensive surveillance are well recognized in LTCFs, where resident turnover and consent procedures often hinder systematic data collection.^{4,36} Generalisability may be further constrained by the heterogeneity of participating LTCFs and their differing case-mix profiles, as well as by the use of a convenience sample, as previously noted in the H4LS study.³⁷ Reliance on manual data collection rather than routinely collected electronic data in LTCFs constrains the scalability of AI-based approaches. This highlights the need for shared data repositories that integrate resident-level information across facilities. Access to such data would enable the application of information extraction methods such as clustering, to classify facilities based on the proportion of residents at high risk for HAIs. Finally, although formal statistical testing of associations between comorbidities and HAIs was not feasible with the present data set, and the observed cluster-specific infection profiles should therefore be considered as hypothesis-generating rather than confirmatory, these patterns suggest that preventive strategies may benefit from being tailored to residents' functional and clinical profiles.

In conclusion, this study indicates that AI-based clustering is feasible in LTCFs and may complement traditional infection risk assessment methods. In a field where AI applications remain limited, these findings support further investigation of machine-learning approaches using larger and more comprehensive data sets. By capturing infection-specific risk profiles that extend beyond global frailty, ML-based clustering highlights subgroups with higher HAI incidence and may inform targeted prevention and more efficient resource allocation. Future research should assess the reproducibility of these results across diverse LTCF settings and evaluate the impact of AI-driven risk stratification on resident outcomes, healthcare costs, and infection prevention strategies.

Supplementary material. The supplementary material for this article can be found at <https://doi.org/10.1017/ice.2026.10413>.

Data availability statement. Pseudonymised case-based data or aggregate data are available upon a data request for research purposes to the ECDC (<https://www.ecdc.europa.eu/en/publications-data/request-tessy-data-research>). Following the approval of the data request, the relevant data can be shared through a secure online platform. The data can be made available for a minimum of 10 years from the end of the study.

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